DO NOW — RECALL

Lesson 049 · Stack<T>: LIFO and Program State

Name: _____ Class: _____ Date: _____

1. How does List<T>.RemoveAt differ from Remove?

2. What is a stack's removal order?

3. What is an instance method?

4. An empty stack executes push(A), push(B), push(C), x=pop(), push(D), push(x), then pop(). Trace the final stack from bottom to top.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

By index versus by matching value.

2. Model answer

LIFO.

3. Model answer

Method called on a specific object.

4. Model answer

A, B, D. x becomes C; pushing x restores C to the top, and the final pop removes C.

By the end, you can…

• A stack is last-in, first-out (LIFO). Push places a value on top, Pop removes the top, and Peek reads without removing.

• A stack can support undo histories and recursive call frames. The physical call stack is a related example but not this Stack<T> collection.

• Always check Count or otherwise handle an empty-stack exception before Pop/Peek.

Vocabulary — know these terms

LIFO — Last In First Out.

Push — Add to top of stack.

Pop — Remove and return top.

Peek — View top without removing.

NEW LEARNING

Read, annotate and explain

Observe

A stack is last-in, first-out (LIFO). Push places a value on top, Pop removes the top, and Peek reads without removing.

Explain

A stack can support undo histories and recursive call frames. The physical call stack is a related example but not this Stack<T> collection.

Apply

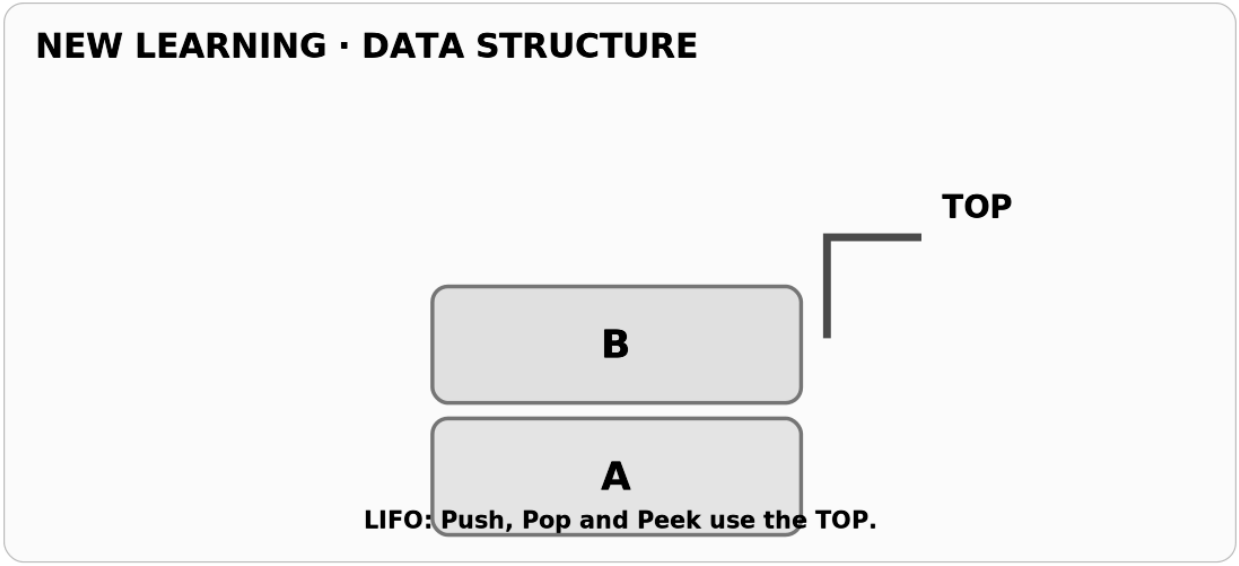
Always check Count or otherwise handle an empty-stack exception before Pop/Peek.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · DATA STRUCTURE: A, B

I DO — WORKED EXAMPLE

Follow the worked method

Trace stack push RED, push BLUE, pop.

Trace and justify

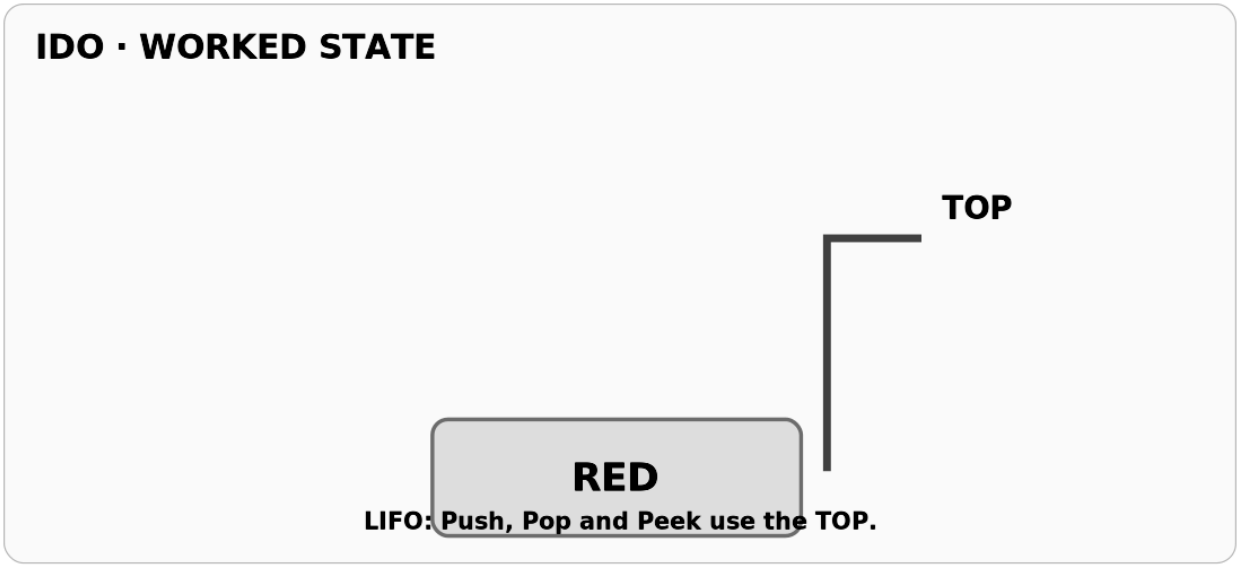
Popped BLUE; remaining stack top RED. Most recent value leaves first.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace stack push RED, push BLUE, pop.

Popped BLUE; remaining stack top RED. Most recent value leaves first.



IDO · WORKED STATE: RED

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Trace a different stack: push 2, push 5, pop, push 9. What is the top?
Source code / input:

```
var s=new Stack<int>();
s.Push(2);s.Push(5);s.Pop();s.Push(9);
```

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Push A, B, C; Pop once, then Peek. State both outputs and remaining stack. [6]

Question 2

Write C# to push 3,6,9 and safely pop one value. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Explain why stack suits undo history but queue suits printing jobs; include two operations. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Peek permanently removes the last value from a stack.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

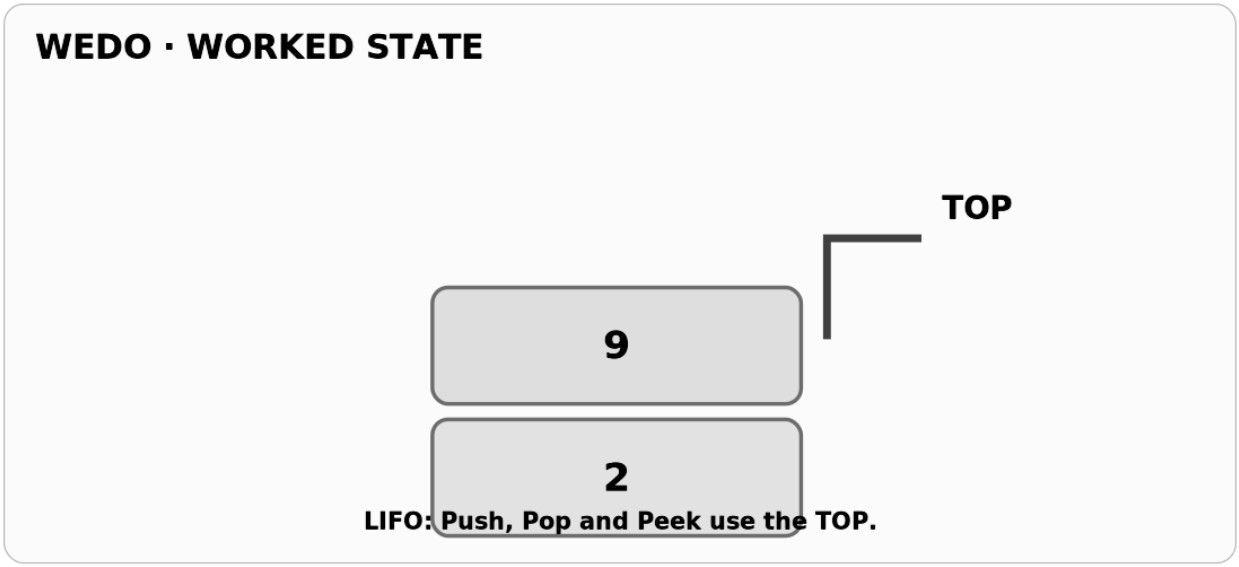
Trace and justify
Stack bottom-to-top [2,9], Peek returns 9; popped value was 5.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a different stack: push 2, push 5, pop, push 9. What is the top?

Stack bottom-to-top [2,9], Peek returns 9; popped value was 5.



WEDO · WORKED STATE: 2, 9